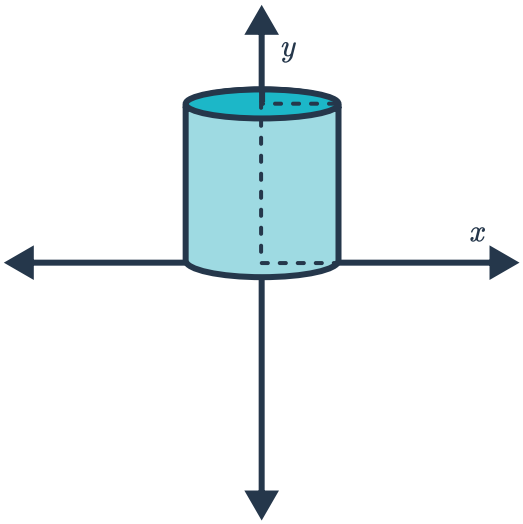
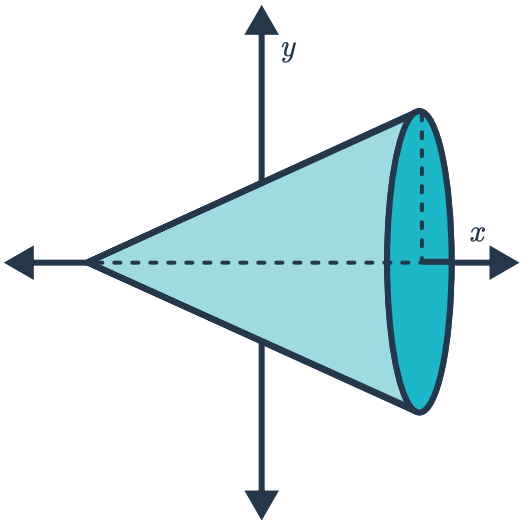
Solids of revolution

1 Identify the axis of rotation for each solid of revolution:

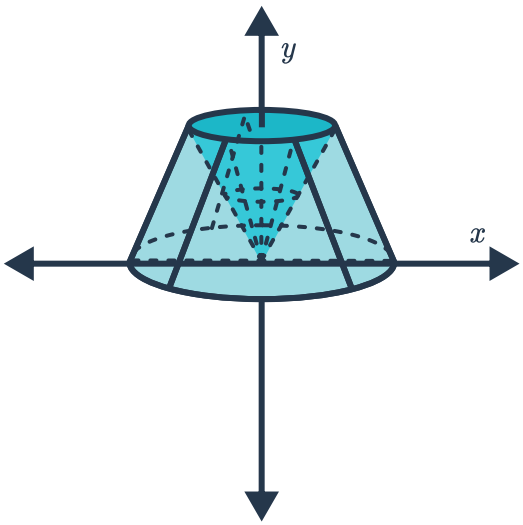
a



b



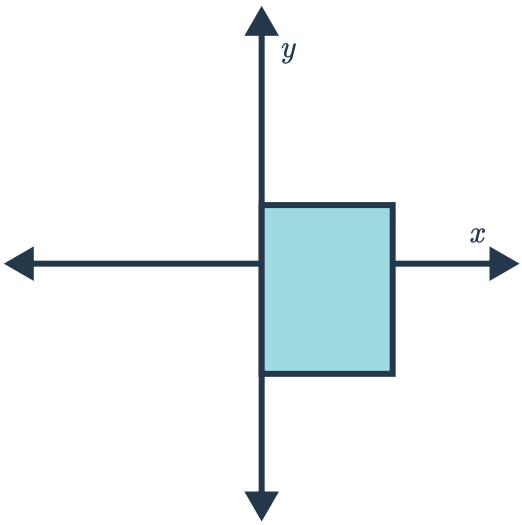
c



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2 Match each figure with its solid of revolution after rotating around the y-axis:

Figure 1



Solid 1

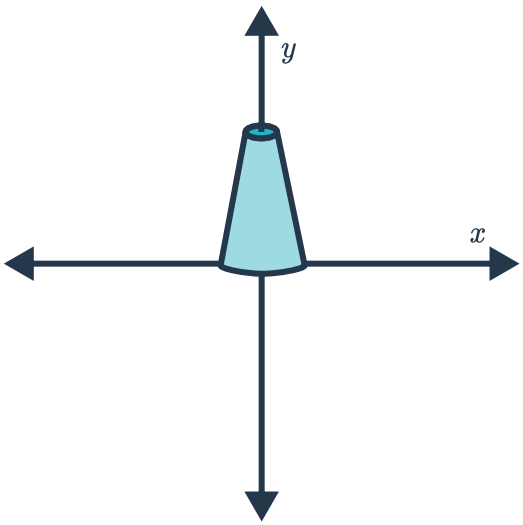
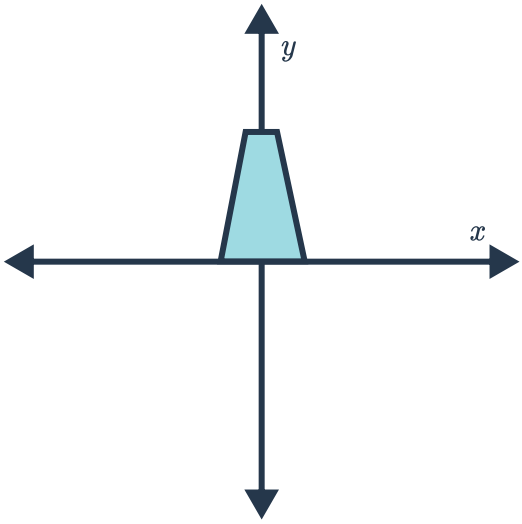


Figure 2



Solid 2

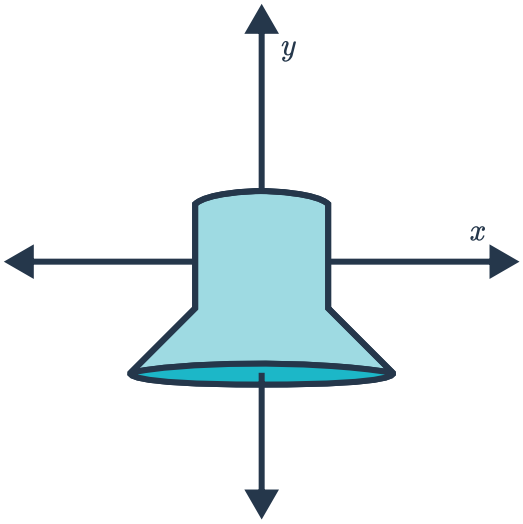
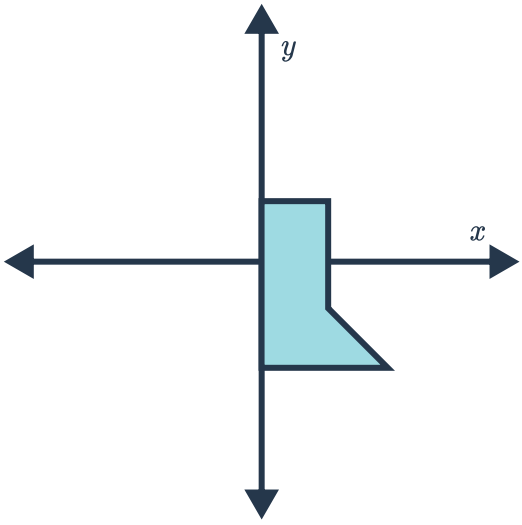
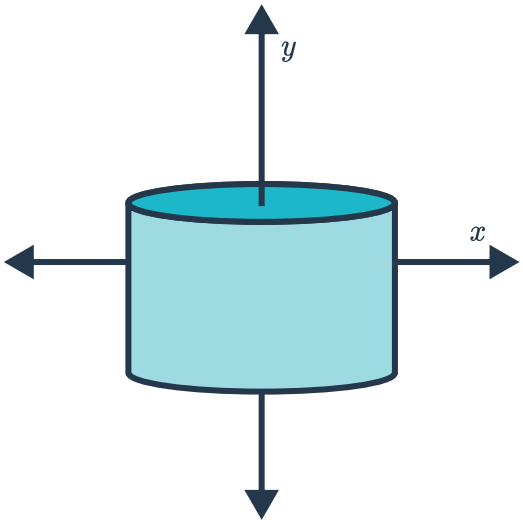


Figure 3



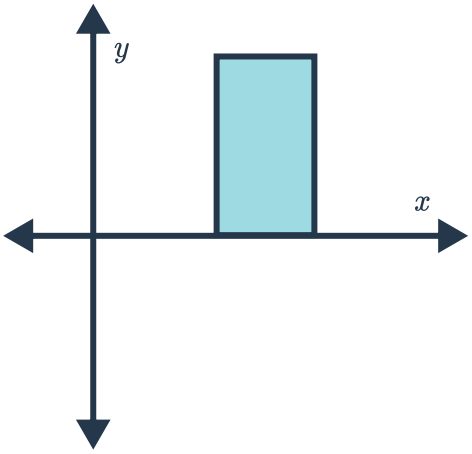
Solid 3



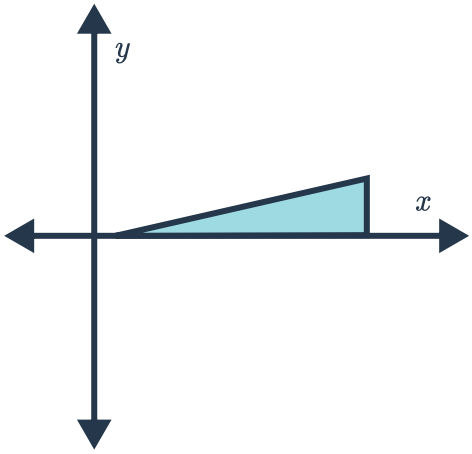
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3 For each of the following figures, sketch the solid of revolution produced by rotating around the x-axis:

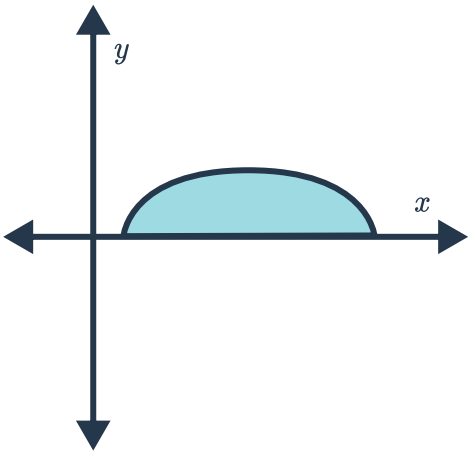
a



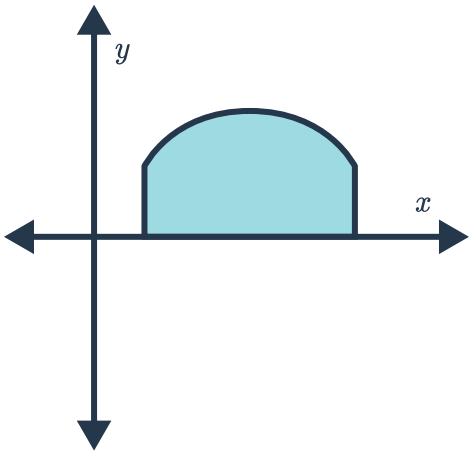
b



c



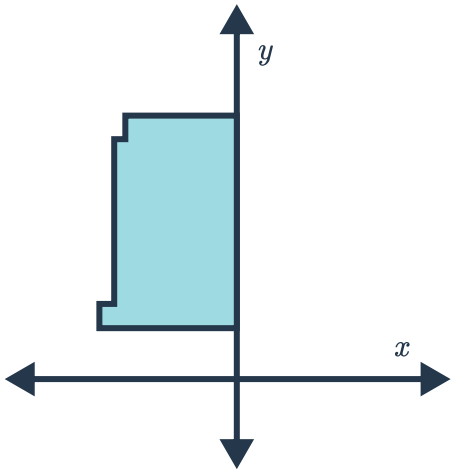
d



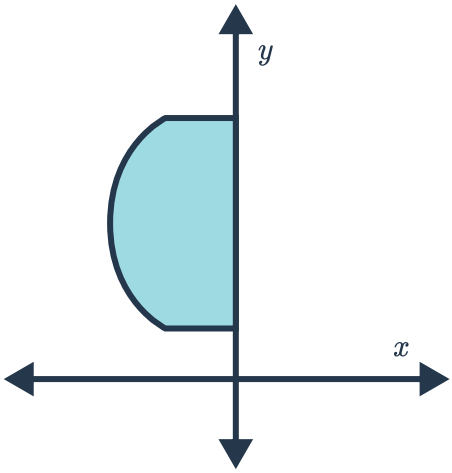
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4 For each of the following figures, sketch the solid of revolution produced by rotating around the y-axis:

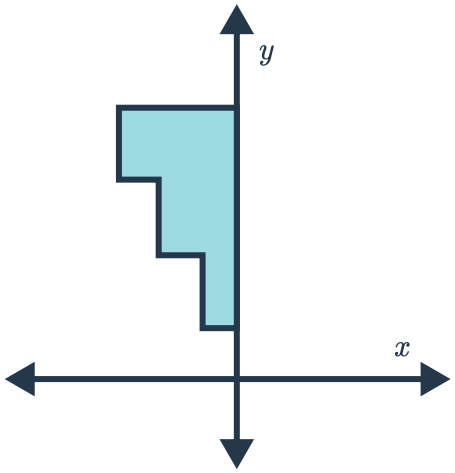
a



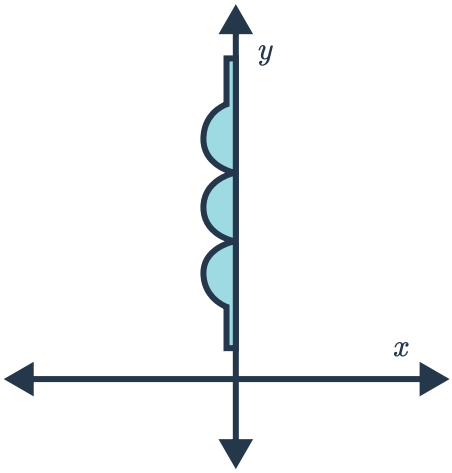
b



c



d



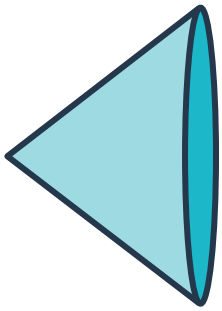
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5 Sketch the two-dimensional figure that produced each solid of revolution around the x-axis:

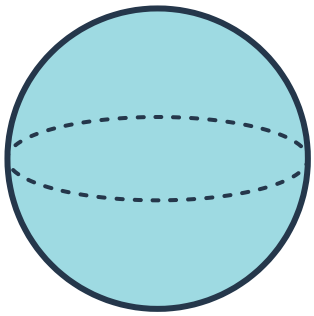
a



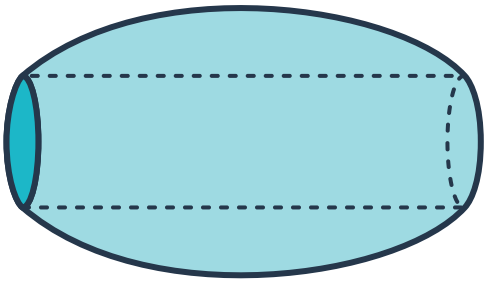
b



c



d



6

a Sketch a two-dimensional figure on the coordinate plane that produces the same solid of revolution when rotated around the x and y-axes.

b Sketch the solid of revolution produced by rotating your figure around the y-axis.

c Sketch the solid of revolution produced by rotating your figure around the x-axis.

7

a Sketch a two-dimensional figure on the coordinate plane that produces different solids of revolution when rotated around the x and y-axes.

b Sketch the solid of revolution produced by rotating around the y-axis.

c Sketch the solid of revolution produced by rotating around the x-axis.

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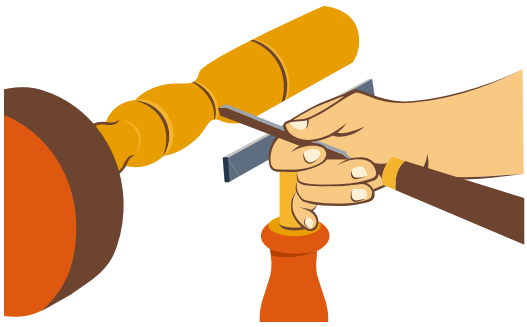
Applications

8 Francesca is wanting to 3D a print bowl. She has worked out that the best way to do this is to create a two-dimensional figure and then rotate it around the y-axis.

a Draw a template that Francesca can use to create a bowl.

b Sketch what your bell will look like when finished.

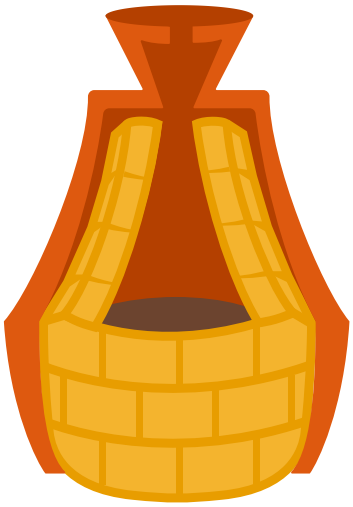
9 A wood turner has been making a goblet (wine glass) on their lathe.
If the lathe is rotating horizontally, draw a figure that when rotated will produce the goblet.



10 Gunner works in a foundry making large bells. The easiest way to do this is by hollowing the required shape out of a block of molding clay. Gunner must construct a template that when rotated will create the hollow.

a Design a template Gunner could use to construct the bell.

b Sketch what your bell will look like when finished.



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